

LLM & GENAI SYSTEMS

LLMs

LoRA / QLoRA

PEFT

Unsloth

vLLM

Transformers

Fine-tuning

Quantization

Model Evaluation

LLM Inference

GPU & DEEP LEARNING

PyTorch

TensorFlow

NVIDIA GPU Systems

RAPIDS (cuDF/CuPy)

Computer Vision

Autoencoders

Distributed Training

MLOPS & PLATFORMS

Cloud & Orchestration

GCP Vertex AI

GKE

Dataflow

GCS

Azure Machine Learning

Deployment & MLOps

Kubernetes

Helm

Istio

Docker

Seldon Core

MLflow

Model Registry

Observability & CI/CD

Grafana

Prometheus

Cloud Build

ArgoCD

Azure DevOps

LANGUAGES & ARCHITECTURE

Python

SQL

Distributed Systems

Software Architecture

Low-latency Systems

Testing & Debugging

CERTIFICATIONS

Oracle Cloud Infrastructure 2024
Generative AI Certified Professional

Microsoft Certified
Azure AI Engineer Associate

Microsoft Certified
Azure Data Scientist Associate

EDUCATION

B.E. Electrical & Electronics

Andhra University, Visakhapatnam

2012 - 2016 · 80%

Lead AI and ML engineer and data scientist with **10+ years** of experience, building production AI/ML/LLM systems. Leads end-to-end AI delivery, combining hands-on model development with technical leadership to turn business problems into scalable, high-throughput, low-latency production systems. Experience spans predictive modelling, forecasting, operations research, computer vision, and generative AI across retail, cybersecurity, manufacturing, insurance, and supply-chain optimization. Selected contributions include modelling on **billion-row security datasets**, supply-chain optimization generating **~\$2.5M** in savings, and self-hosted LLMs saving an estimated **~\$300K** annually while improving availability during peak demand.

WORK EXPERIENCE

- Lead Artificial Intelligence Engineer

Best Buy · Bengaluru, India

Jul 2025 - Present

- Architected a config-driven Python LoRA/QLoRA fine-tuning and evaluation framework using Unsloth, PEFT, Transformers, and Accelerate; containerized vLLM on GKE with autoscaling, Istio, blue-green releases, and observability supported estimated annual API-cost savings of **~\$300K**. Kept model inputs company-managed and reduced external API exposure during peak demand.
 - Built fraud systems across batch, streaming, and real-time models. Beam/Dataflow feature pipelines handled **40,000 events/sec** at peak; supported a **200 ms scoring SLA** with secure, monitored GKE/Seldon Core inference and load testing.
 - Lead architecture and production delivery across search, personalization, fraud, and cybersecurity; partner with data scientists to turn validated models into secure, monitored services.
- Lead ML Engineer - ML Engineering & Operations

Wipro · Client: Major US Retail Corporation

Aug 2023 - Jul 2025

- Led ML/MLOps engineers and Morpheus/RAPIDS model lifecycles on **NVIDIA DGX (8×A100)**, training per-entity autoencoders on **billion-row CloudTrail datasets** with MLflow model management, continuous evaluation, debugging, and Grafana alerting. Completed **300+ résumé screens** and **50+ technical interviews** at Wipro.
 - Launched GPU-accelerated phishing classification at **2,400 emails/sec - 2× prior throughput** and computer-vision/OCR abuse-inbox triage that doubled case volume, integrating predictions with Microsoft Sentinel and security-response workflows.
- Senior Data Scientist — Data Science / Azure ML

Accenture

May 2021 - Sep 2023

- Led cross-functional architecture and launch of an Azure ML quality-prediction product: **\$2.5M annual savings**, inspections **130 → 160/day**, **+17%** defect detection, and **+20%** customer approval.
 - Operationalized data preparation, optimization, retraining, monitoring, and CI/CD with Azure ML Studio, Function Apps, Logic Apps, and Azure DevOps; mentored engineers and coordinated business, data science, platform, and client stakeholders.

GPU & DISTRIBUTED
COMPUTING

CUDA GPU Libraries

GPU-Accelerated Computing

OPTIMISATION & SIMULATION

Google OR-Tools

Vehicle Routing

Mixed-Integer Programming

Constraint Modelling

Operations Research

Supply-Chain Optimization

MODEL LIFECYCLE

Experiment Tracking

Automated Retraining

Vertex AI Pipelines

Azure ML Pipelines

Vertex AI Feature Store

CLOUD SERVICES

Cloud Run

Cloud Functions

Azure Function Apps

Logic Apps

Google KCC

PRODUCTION ENGINEERING

ML System Architecture

Cloud-Native ML Systems

GitOps

JWT Authentication

Load Testing

Unit / Integration Testing

LANGUAGES & DATA

BigQuery

MS SQL Server

Redis Caching

Azure Blob Storage

Artifact Registry

Pandas / NumPy

Dask

Git

Linux / Shell Scripting

ENGINEERING LEADERSHIP

Architecture Reviews

Technical Mentoring

Technical Hiring

Team Building

Stakeholder Alignment

Cross-Functional Delivery

EARLIER EXPERIENCE

Data Scientist

GD Research Centre Pvt Ltd

May 2020 - May 2021

Designed and deployed Python forecasting pipelines across 15+ industry verticals, from heterogeneous data preparation through client-facing results.

Deep Learning Engineer

Tata Consultancy Services

Jul 2019 - Mar 2020

Built and deployed an end-to-end semiconductor ML system: distributed training on **~1 million images in ~3 hours; 90% classification accuracy**. Deployed at Amkor, South Korea, and trained the client team.

Senior Software Engineer - ML
Infosys

May 2016 - Jul 2019

Delivered design, coding, unit/integration testing, and production support; transitioned to Python ML in 2018 and shipped lead-conversion prediction.

PROJECTS

LLM Fine-tuning & Serving Framework

Best Buy · Jul 2025 - Present

Config-driven Python platform for LoRA/QLoRA fine-tuning, data preparation, multi-task training, automated evaluation, artifact management, and vLLM serving. Supports Unsloth and PEFT/Transformers with Accelerate; adapters hot-load into a shared GKE deployment.

~\$300K estimated annual API-cost savings

Real-time Retail Fraud ML Platform

Best Buy · Aug 2025 - Jan 2026

Large-scale Beam/Dataflow pipelines transform clickstream and transaction events into Vertex AI Feature Store signals. XGBoost models deploy through Seldon Core on GKE with JWT security, load testing, GitOps releases, and production monitoring.

Apache Beam

Vertex AI

40k feature events/sec · 200 ms scoring SLA · Full lifecycle deployment

AWS Account Behavioral Fingerprinting

Wipro · Jan 2024 - Aug 2025

Per-entity autoencoders learn normal AWS account, user, service, and machine activity. Morpheus/RAPIDS training and inference run on DGX (8xA100) over billion-row CloudTrail datasets with continuous retraining and Grafana alerts.

RAPIDS / cuDF

Billion-row datasets · Continuous model lifecycle · 8xA100 DGX

Semiconductor Computer Vision Platform

TCS / Amkor Technology · Jul 2019 - Mar 2020

Built and deployed an end-to-end semiconductor ML system: distributed training on **~1 million images in ~3 hours**, with **90% classification accuracy**. Deployed at Amkor, South Korea, and trained the client team.

DenseNet

Distributed GPU

~1M images · ~3-hour training cycles · Production launch in South Korea

Proofpoint Phishing Classification

Wipro · Oct 2023 - Feb 2024

Built Morpheus training and inference pipelines on DGX for preprocessing, feature engineering, model evaluation, and live scoring. Integrated security-team feedback and Microsoft Sentinel data to catch phishing that bypassed initial filters.

RAPIDS

DGX

Microsoft Sentinel

2,400 emails/sec · 2x prior throughput · Production security workflow

Azure ML Product Quality Platform

Accenture / BamaGruppen · May 2021 - Dec 2021

Led an end-to-end ML product from data preparation and model optimization through automated retraining, CI/CD, monitoring, client demonstrations, and ongoing production support on Azure.

Azure ML

Logic Apps

CI/CD

Monitoring

\$2.5M/year · +17% detection · +20% approval · Launched product

RESEARCH-TO-PRODUCTION EVIDENCE

vLLM Model Chooser

Live open-source tool

Matches 115 vLLM-compatible models to GPU VRAM, quantization, and KV-cache constraints across A100 through B200.

Local AI Agent (MLX)

Open-source project

Runs LLMs fully locally on Apple Silicon with a single command, with no cloud dependency or account requirement.

Technical Direction

Research, Platform & Product Teams

Turns experimental models into evaluated, monitored, low-latency services while aligning data science, platform, security, and product stakeholders.